

COP2073C Module 9 Programming Assignment

Instructions

Using the mtcars dataset included in the base R distribution, create a model of fuel efficiency, measured in miles per gallon (MPG), as a function of vehicle weight (in thousands of pounds).

Write an R script that performs the following tasks:

1. **Plot the Data:**
 - Create a scatter plot of MPG (miles per gallon) on the y-axis and wt (vehicle weight in thousands of pounds) on the x-axis using base R plotting functions.
 - Provide an appropriate title, labels for both axes, and ensure that the plot is visually clear and well-labeled.
2. **Fit a Linear Regression Model:**
 - Use the lm function to fit a linear regression model with mpg as the dependent variable and wt as the independent variable. Use formula notation in the model.
3. **Add the Fitted Regression Line:**
 - Add the fitted regression line to the scatter plot created in Step 1 using the abline function.
4. **Interpret the Slope:**
 - Interpret the estimated slope of the regression line.
 - Determine if the slope indicates a statistically significant relationship between wt and mpg. Support your answer with appropriate statistical evidence, such as the p-value and confidence interval, by running the summary and confint functions. Use a commonly accepted significance threshold (e.g., a p-value of 0.05).
5. **Predict MPG for a Given Vehicle Weight:**
 - Convert the vehicle weight to thousands of pounds (e.g., 6,000 lbs = 6).
 - Produce a point estimate and a 95% prediction interval for the MPG of a car that weighs 6,000 pounds. Create a new data frame, and then use the predict function with the original model and the new data frame to generate the prediction.
6. **Print the Prediction:**
 - Print the predicted MPG value along with the 95% prediction interval.
7. **Evaluate the Model's Prediction:**
 - Evaluate whether you trust the model's predictions for a vehicle of this weight. Discuss whether the weight is within the range of the dataset and whether there are reasons to question the model's accuracy for this prediction.

Non-Functional Requirements:

- Include a 4-line ID header at the beginning of your script.
- Include vertical spacing (a blank line) between logical blocks for readability.
- Comment your code thoughtfully (avoid excessive commenting).
- Ensure each line of code does not exceed 80 column

Sample Output:

Slope Estimate: -5.344472

p-value: 1.293959e-10

95% Confidence Interval for Slope: -6.486308 -4.202635

The slope is statistically significant at the 0.05 level, indicating a significant relationship between weight and fuel efficiency.

Predicted MPG for a 6,000-pound vehicle:

	fit	lwr	upr
1	5.218297	-1.85279	12.28938

Note: The weight value is outside the range of the dataset, which limits the accuracy of the prediction. Use caution when interpreting this result.